

TAGOOR PARAMATHMA JAYAMANGALA

AI & Data Engineer

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[LinkedIn](#) | [Portfolio](#)

PROFILE SUMMARY

Data Engineer and Generative AI application developer currently building and deploying production LLM-powered systems at Welspun Transformation Services Limited (WTSL), part of the Welspun World conglomerate. Skilled in Python, SQL, and PostgreSQL, with proven work in data modeling, schema design, ETL and data-pipeline orchestration (Apache Airflow, Cron), database migrations (Alembic), and REST / async API development using FastAPI, LangGraph, and LangChain. Built multilingual conversational AI assistants and real-time chatbot backends powered by agentic AI workflows and Retrieval-Augmented Generation (RAG). Holds a globally recognized Diploma in Data Analytics from SUNY Potsdam (USA) and is NASSCOM-recognized for Python, combining strong data fundamentals with hands-on Generative AI engineering.

TECHNICAL SKILLS

Programming & Scripting : Python (NumPy, Pandas, Matplotlib, Seaborn, Scikit-learn, RapidFuzz, fpdf2), SQL, Async Python

Generative AI & LLM Engineering : LLM Integration (Azure OpenAI, Groq / Llama 3), LangGraph, LangChain, Retrieval-Augmented Generation (RAG), Agentic AI & ReAct Agents, Prompt Engineering, Tool-Calling Workflows, Conversational AI

Backend & API Engineering : FastAPI, REST & Async APIs, WebSockets, SQLAlchemy, Alembic, Pydantic, JWT & OTP Authentication, Webhook Integrations (WhatsApp / WATI)

Data Engineering : Data Modeling, Relational Schema Design, Multi-Source Data Integration, ETL / Data Pipelines, Pipeline Orchestration, Scheduling & Monitoring (Apache Airflow, Cron), Database Migrations (Alembic), Connection Pooling, Query Optimization, Automated Email Reporting (SMTP)

Databases : PostgreSQL, MySQL, MongoDB, SQL

DevOps & Deployment : Docker, Docker Compose, Nginx, Gunicorn / Uvicorn, systemd, Git, Prometheus, Pytest, SSH Tunneling (Paramiko)

BI & Visualization : Power BI (DAX, Power Query), Tableau, Google Looker Studio, Streamlit, Plotly, Advanced Excel

Data Analysis : Data Cleaning, Exploratory Data Analysis (EDA), Data Preprocessing, Statistical Analysis

EXPERIENCE

Data Engineering Trainee

July 2025 – Present

Welspun Transformation Services Limited (WTSL) — Welspun World

Key Projects delivered in this role:

► **Project: WelHire Analytics KPI ETL — Multi-Source Data Warehouse & Reporting Pipeline** — *Two-Layer Batch Analytics Engine*

Tech Stack: Python, Pandas, NumPy, Apache Airflow (DockerOperator), PostgreSQL (psycopg3), MongoDB (PyMongo), MySQL (PyMySQL), SQLAlchemy, Docker, Cron, SMTP, Paramiko / SSH Tunnel

- Built a two-layer batch ETL pipeline that consolidates recruitment data from 6 PostgreSQL databases and MongoDB into a unified analytics warehouse, then computes 11 business KPIs for recruiter, tenant, and candidate reporting.
- Built Layer 2 as an independent, registry-driven KPI engine (interview funnel stages, attempt and completion rates, candidate ratings, role- and tenant-level breakdowns) with per-KPI replace / append load modes; optimized data access using centralized singleton connections and SQLAlchemy connection pooling (roughly 70% fewer DB connections) plus column-level loading for memory efficiency.
- Orchestrated the end-to-end Layer 1 to Layer 2 flow through a single executor with retry logic, file and console logging, and automated SMTP email run-reports; scheduled it daily via an Apache Airflow DockerOperator DAG (credentials mounted at runtime, never baked into the image), with cron as a fallback.

- **Business & Industry Impact:** Gives recruitment and leadership teams a single, automated source of daily hiring KPIs across tenants and recruiters — replacing manual report pulls and enabling data-driven decisions on funnel health and platform usage.

► **Project: Candidate Live-Status Tracker ETL — Real-Time Interview Funnel Pipeline** — *Per-Minute Multi-Source Pipeline & Live Dashboard*

Tech Stack: Python, Apache Airflow, PostgreSQL, MongoDB (PyMongo), MySQL (mysql-connector-python), SQLAlchemy, Streamlit, Plotly, Docker / Docker Compose, Pytest, Cron

- Built a near real-time, per-minute ETL pipeline that fuses live state from three read-only source systems — PostgreSQL scheduled interviews and two MongoDB event collections — into a single MySQL analytics table showing exactly where every candidate sits in the interview funnel.
- Modeled the funnel as 14 ordered stages spanning email-sent, device and network checks, in-progress interviewing, completion, and post-processing, across two interview engines (Classic and Agentic), with deterministic stage-resolution logic and a composite (candidate_id, jd_id) primary key reconciled by upsert plus history append on every tick.
- Orchestrated the pipeline via an Apache Airflow DAG running every minute and shipped a live Streamlit + Plotly dashboard for funnel, stuck-candidate, and ETL-health monitoring; packaged the full stack (Airflow, migration runner, dashboard) with Docker Compose for identical local and production runs.
- **Business & Industry Impact:** Gives recruiters and operations a live view of candidate progress and bottlenecks during interviews — surfacing stuck or stalled candidates in real time and cutting reliance on manual status checks.

► **Project: ChittiGPT — Enterprise Employee AI Assistant (Welspun Group)** — *HR & Self-Service Copilot*

Tech Stack: Python, FastAPI, LangGraph, WebSockets, Azure OpenAI (GPT-4.1-mini), RAG, PostgreSQL, SQLAlchemy, Alembic, JWT, Prometheus, Docker

- Built the backend for ChittiGPT, an internal AI assistant for Welspun Group employees that handles everyday HR tasks through chat — leave, attendance corrections, leave balance, payslips, document downloads (Form 16, mediclaim e-card), IT/HR tickets, candidate referrals, and policy and insurance questions.
- Built the AI agent (LangGraph + Azure OpenAI) connecting 20+ company services as tools, plus a knowledge-base search (RAG) that answers HR, policy, and insurance questions directly from official company documents.
- Implemented secure access (JWT + OTP email login and single sign-on), a PostgreSQL database (15+ versioned updates), live system monitoring (Prometheus), and Docker deployment.
- **Business & Industry Impact:** Gives Welspun employees one chat assistant for routine HR and self-service tasks — reducing dependence on HR teams and portals and speeding up everyday requests, in line with the enterprise shift toward AI copilots for employee self-service.

► **Project: Sintex — Multilingual WhatsApp AI Assistant for Plumbers** — *Loyalty, Rewards & Support Agent*

Tech Stack: Python, FastAPI, LangGraph, LangChain, Azure OpenAI (GPT), PostgreSQL, SQLAlchemy, Alembic, RAG, rapidfuzz, WATI (WhatsApp API), fpdf2, Docker

- Built a WhatsApp chatbot that lets plumbers talk to Sintex in their own language (English, Hindi, Marathi, or Hinglish) to check and redeem reward points, claim points by scanning product QR codes, raise support tickets, find dealers, and ask product questions.
- Built the AI agent (LangGraph + Azure OpenAI) with 8 connected features, including an assistant that answers product questions directly from official Sintex documents (RAG).
- Made the reward-claim process fully reliable by running it through fixed, rule-based logic instead of the AI — so the recorded data is always correct and the AI can never enter wrong values.
- **Business & Industry Impact:** Brings the loyalty program into the app plumbers already use, making it effortless to earn and redeem rewards in their own language — improving engagement and reducing manual support.

Data Analytics Intern

360DigiTMG

- Gained hands-on experience in Advanced Excel, Python, SQL, and data visualization tools; performed EDA, statistical analysis, and data pre-processing on real-world projects.
- Developed reports and dynamic dashboards using Power BI, Tableau, and Google Looker Studio to visualize insights for stakeholders.

Machine Learning Intern

Henotic Technologies Pvt. Ltd.

- Conducted Exploratory Data Analysis (EDA) on real-world datasets and handled missing values, outliers, and skewness using appropriate preprocessing techniques.
- Developed and tested machine-learning models using Python.

EDUCATION

B.Tech in Electrical & Electronics Engineering <i>S.R.K.R Engineering College CGPA: 7.93</i>	2020 – 2024
Intermediate (MPC) <i>Sri Chaitanya Junior College CGPA: 9.75</i>	2018 – 2020
SSC <i>Bashyam E.M Public School CGPA: 9.8</i>	2017 – 2018

CERTIFICATIONS & TRAINING

- Diploma in Data Analytics, SUNY Potsdam (with NEF & 360DigiTMG)
- Python for Data Science, NASSCOM & 360DigiTMG (Gold Certificate)
- Professional Data Analytics Course Completion, 360DigiTMG
- SQL for Data Analysis, Advanced Excel, Power BI, Tableau — 360DigiTMG

ACHIEVEMENTS

- Awarded Diploma in Data Analytics by SUNY Potsdam, USA with 86% under the prestigious Appuji Scholarship.
- Secured Gold Certificate in the Python Fundamentals exam conducted by NASSCOM.
- Secured 2nd position at Projects Expo 2024 for the academic project in the EEE Department.
- Led a team of 6 in developing a Solar-Powered Wireless Charging System for EVs.

PERSONAL SKILLS

High attention to detail · Strong analytical & problem-solving skills · Effective communication & teamwork · Excellent time management & adaptability